INCH-POUND
MIL-PRF-1/501C
27 January 1999
SUPERSEDING
MIL-E-1/501B
26 April 1972

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 3KP1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Ehk	Alt
Unit:	V	V dc	V dc	V dc	V dc	fL	Meg	Meg	V dc	ft
Maximum:	6.9	-200	550	1,100	2,750	- - -	1.5	1.0	±125	10,000
Minimum:	5.7	0	- - -	- - -	550 note 2	4.0	- - -	- - -	- - -	- - -
Test conditions:	6.3	Adj	- - -	Focus	1,500	- - -	- - -	- - -	- - -	- - -

GENERAL:

Qualification - Required.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE 1. Testing and inspection.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Units
Qualification inspection							
Base material insulating quality	1216	- - -	Zone 5 (minimum)	- - -	- - -	- - -	- - -
Pressure (implosion)	1141	- - -		- - -	- - -	- - -	- - -
Vibration	5111	- - -		Width	- - -	1.0	mm
Direct-interelectrode capacitance	1331	- - -	g1 to all D1 to D2 D3 to D4 D1 to all D2 to all D3 to all D4 to all	Cg1 C1D2 C3D4 CD1 CD2 CD3 CD4	- -	11.0 4.0 4.0 15.0 11.0 10.0 11.0	pF pF pF pF pF pF pF
Deflection-factor uniformity	5248	- - -		- - -	- - -	- - -	- - -
Focusing voltage, zero-bias	5246	- - -		Eb1	240	450	V dc
Neck and bulb alignment (electrostatic types)	5101	- - -		Diam	- - -	1.87	Inches
Cathode illumination	5216	- - -		- - -	- - -	- - -	- - -
Conformance inspection, part 1							
Bulb, screen and faceplate quality	5106	- - -		- - -	- - -	- - -	- - -
Voltage breakdown	5201	- - -		- - -	- - -	- - -	- - -
Voltage breakdown (electrostatic types)	5201	- - -		- - -	- - -	- - -	- - -
Gas “cross”	5206	1	Light = 4 fL	- - -	- - -	- - -	- - -
Light output	5221	1		Light	4.0	- - -	fL
Modulation	5223	- - -	Light = 4 fL	ΔEc1	- - -	45	V dc
Spot displacement (leakage)	5231	- - -		Displ	- - -	7.0	mm
Spot position (electrostatic deflection)	5231	- - -		- - -	- - -	15.0	mm
Grid cutoff voltage	5241	- - -		Ec1	- - -	-67.5	V dc
Grid No. 1 leakage current	5251	- - -		- - -	- - -	- - -	- - -
Anode No. 2 leakage current	5251	- - -		- - -	- - -	- - -	- - -

See notes at end of table I.

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TABLE 1. Testing and inspection -Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Units
Conformance inspection, part 2							
Heater current	1301	- - -		If	540	660	mA
Electrode current (anode No. 1)	5201	- - -	Ec1 = 0	Ib1	-15	10	μA dc
Electrode current (cathode)	5201	- - -	Light = 4 fL	Ik	- - -	1,000	μA dc
Base alignment (electrostatic types)	5101	- - -	+3D4; pin No. 1	- - -	- - -	- - -	- - -
Angle between traces	5101	- - -		Angle	87	93	Degrees
Neck and base alignment (electrostatic types)	5101	- - -		- - -	- - -	- - -	- - -
Stray light emission (conventional types)	5216	- - -		- - -	- - -	- - -	- - -
Line width A (electrostatic types)	5226	- - -	Light = 4 fL	Width	- - -	0.4	mm
Line width B (electrostatic types)	5226	- - -	Light = 4 fL	Width	- - -	0.4	mm
Focusing voltage	- - -	- - -	Light = 4 fL	Eb1	240	450	V dc
Deflection factor	5248	- - -	1D2	DF	75	102	V dc/in.
Deflection factor	5248	- - -	3D4	DF	57	78	V dc/in.
Heater-cathode leakage current	5251	- - -		- - -	- - -	- - -	- - -
Secureness of base, cap and insert	1101	- - -		- - -	- - -	- - -	- - -
Base pin solder depth	1111	- - -		- - -	- - -	- - -	- - -
Permanence of marking	1105	- - -		- - -	- - -	- - -	- - -
Conformance inspection, part 3							
Life test	- - -	3	Group C; Light = 4 fL; Eb2 = 2,750 V dc; t = 500 hours (min)	- - -	- - -	- - -	- - -
Life-test end points:	- - -						
Line width A		- - -	Light = 3.4 fL	Width	- - -	0.4	mm
Line width B		- - -	Light = 3.4 fL	Width	- - -	0.4	mm
Modulation		- - -	Light = 3.4 fL	ΔEc1	- - -	45	V dc

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. This value is recommended only for low-velocity deflecting and low-ambient light levels.
3. During life-test operation the heater voltage shall be regularly interrupted on a 13 minutes “on”, 7 minutes “off” cycle. Life test minimum duration shall be the total of “on” plus “off” hours.

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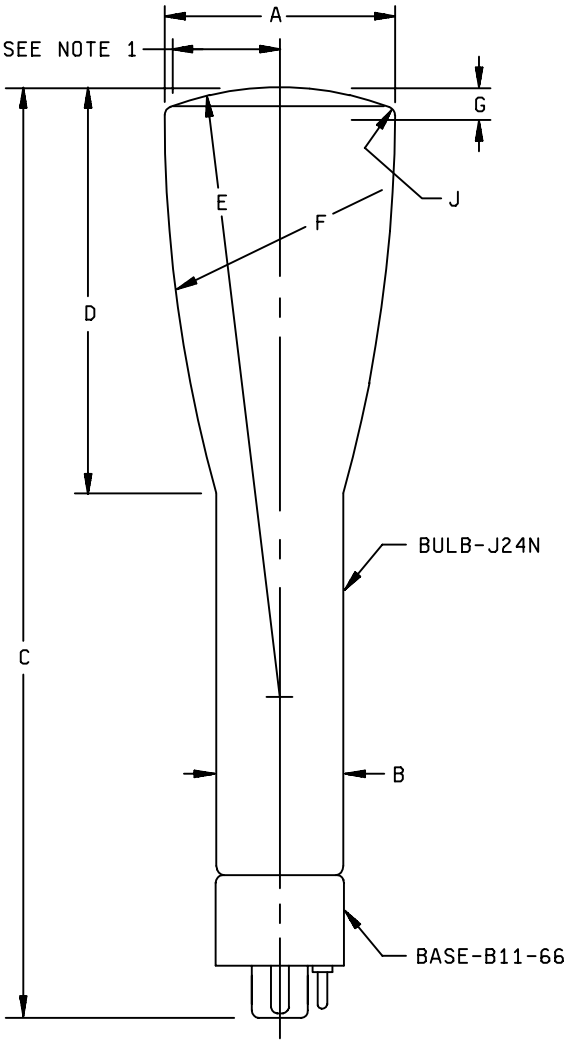
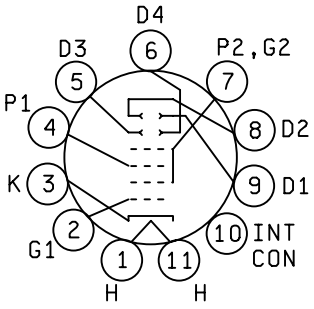


Dimensions in inches with metric equivalents (mm) in parentheses		
Ltr	Minimum	Maximum
Qualification inspection		
	Base: B11-66	
Conformance inspection, part 2		
A	2.937 (74.60)	3.063 (77.80)
B	1.560 (39.62)	1.680 (42.67)
C	11.250 (285.75)	11.750 (298.45)
Reference dimensions (see note 2)		
D	4.875 (123.83)	
E	8.000 (203.20) Rad.	
F	15.000 (381.00) Rad.	
G	0.350 (8.89)	
J	0.250 (6.35) Rad.	

NOTES:

1. The minimum useful screen radius shall not be less than 1.375 (34.93 mm).
2. These dimensions are for information only and are not required for inspection purposes.



BASING DIAGRAM

FIGURE 1. Outline drawing of electron tube type 3KP1.

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Custodians:
Army - CR
Navy - EC
Air Force - 80

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 11, 17

Preparing activity:
DLA - CC

(Project 5960-3467-04)

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